



RS004 Week 15 — English Worksheet

Topic: Multi-Level Structure · **Course:** Platformer Game · **Time:** about 45 minutes

This week your game holds **many levels**. Every level is a dictionary of platforms, coins, enemies, and a finish. A `load_level()` function sets up the current one, and reaching the finish loads the **next** — until the last one, which means **victory**.

Keep these words handy: **levels list**, `load_level()`, **next level**, **victory**, **game_state**.

1 · Predict 🧠

Read each snippet. Before you run it, write what you think will happen.

```
LEVELS = [
    { "platforms": [...], "coins": [...], "enemies": [...], "finish": (...) },
    { "platforms": [...], "coins": [...], "enemies": [...], "finish": (...) },
]
```

How many levels are in this list? How would you read the first level's coins?


```
def load_level(idx):
    data = LEVELS[idx]
    platforms = [pygame.Rect(*p) for p in data["platforms"]]
    ...
    return platforms, coins, enemies, finish
```

What does `load_level(1)` set up? Why return four things at once?


```
if player.colliderect(finish):  
    current_level += 1  
    if current_level < len(LEVELS):  
        platforms, coins, enemies, finish = load_level(current_level)  
    else:  
        game_state = "victory"
```

The player clears the last level. Which branch runs — load or victory?

2 · Trace the Level Flow 12 34

There are 3 levels. The player keeps reaching the finish. Fill in `current_level` and what happens.

```
start:          current_level = 0
clear level:    current_level = ____ → loads level ____
clear again:    current_level = ____ → loads level ____
clear again:    current_level = ____ → game_state = ____
```


3 · Spot the Bug 🐛

Each snippet is broken. Rewrite it so it works, then explain why the original was wrong.

Bug A — Reaching the finish should load the next level. Right now the player just keeps playing the same level forever.

```
if player.colliderect(finish):  
    platforms, coins, enemies, finish = load_level(current_level)
```

Hint: `current_level` never advances. Increase it first.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — After the last level the game should show victory. Right now it crashes with an index error when it tries to load a level that doesn't exist.

```
if player.colliderect(finish):  
    current_level += 1  
    platforms, coins, enemies, finish = load_level(current_level)
```

Hint: check whether `current_level` is still inside the list before loading.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — `load_level` should turn the data into real game objects. Right now the enemies don't move because they were never made into `Enemy` objects.

```
def load_level(idx):
    data = LEVELS[idx]
    platforms = [pygame.Rect(*p) for p in data["platforms"]]
    coins = [pygame.Rect(c[0], c[1], 20, 20) for c in data["coins"]]
    enemies = data["enemies"]
    finish = pygame.Rect(*data["finish"])
    return platforms, coins, enemies, finish
```

Hint: the enemy tuples need to become `Enemy(...)` objects, like the platforms became `Rect` s.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Author Your Levels

Start from your working multi-level loader.

1. Write the data for your **first** level (platforms, coins, enemies, finish). Reuse a level you already designed.

2. Make your **second** level a little harder than the first. What did you change to raise the difficulty?

3. **Stretch:** make level three the hardest, with a tricky final jump. Describe it.

5 · Build & Show 📷

Build **your own** three levels that play one after another, ending in a victory screen.

When it works, send a **photo or video** clearing all three levels, then explain what you did. Use these starters — write 4 to 6 sentences.

First, I store every level as data in **LEVELS** so that ...

`load_level()` sets up the current level by ...

When the player clears a level, the game ...

After the last level, the game shows ...

One tricky moment was ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video clearing all three levels. Teach it to someone new. Try to use these words: **levels list**, **load_level**, **next level**, **victory**, **game_state**.

1. Show level 1 and clear it.
2. Show the game loading level 2, then 3.
3. Read the "next level or victory" check out loud and explain it.
4. Reach the victory screen.

Write what you will say in your video. Plan it here first.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.